Matrix Multiplication

We want to define the product of two matrices.

- We have defined the product of an $m \times n$ matrix and an $n \times 1$ matrix.

- If A is an $m \times n$ matrix and B is an $n \times p$ matrix, then the product AB is defined and is an $m \times p$ matrix.
- We write
$$B = [\, \mathbf{b}_1 \; \mathbf{b}_2 \; \cdots \; \mathbf{b}_p \,]$$
where $\mathbf{b}_1, ..., \mathbf{b}_p$ are $n \times 1$ vectors (therefore the products $A\mathbf{b}_1, ..., A\mathbf{b}_p$ are meaningful).
- We define
$$AB = [\, A\mathbf{b}_1 \;\; A\mathbf{b}_2 \;\; \cdots \;\; A\mathbf{b}_p \,]$$

$$m \times n \qquad n \times p \qquad \rightsquigarrow \qquad m \times p$$

Matrix Multiplication

$$B = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_p] \quad \Rightarrow \quad AB = [A\mathbf{b}_1 \ \cdots \ A\mathbf{b}_p]$$

Example

$$\underbrace{\begin{bmatrix} 1 & -3 & 4 \\ 2 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 5 & -1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} 2 & -1 \\ 1 & 0 \\ -3 & \frac{1}{2} \end{bmatrix}}_{B=[\mathbf{b}_1 \ \mathbf{b}_2]} = \begin{bmatrix} \underbrace{\begin{bmatrix} 1 & -3 & 4 \\ 2 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 5 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}}_{A\mathbf{b}_1} & \underbrace{\begin{bmatrix} 1 & -3 & 4 \\ 2 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 5 & -1 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ \frac{1}{2} \end{bmatrix}}_{A\mathbf{b}_2} \end{bmatrix}$$

$$= \begin{bmatrix} -13 & 1 \\ 1 & -\frac{3}{2} \\ 7 & -1 \\ 8 & -\frac{1}{2} \end{bmatrix}$$

Example

The product $\begin{bmatrix} 1 & -3 & 4 \\ 2 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 5 & -1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 0 \\ -1 & 0 & 1 \\ 1 & 2 & -1 \\ 0 & 1 & -1 \end{bmatrix}$ is not defined.

$$4 \times 3 \qquad 4 \times 3 \rightsquigarrow \text{☹}$$

- The number of columns of the first matrix is NOT equal to the number of rows of the second matrix.

Remember the Rule :

$$m \times n \quad n \times p \quad \rightsquigarrow \quad m \times p$$

Example.

- If A is 7×4 and B is 4×3 then AB is defined and is a 7×3 matrix.
- However, if A is 3×2 and B is 5×2 then AB is not defined!

Another method of calculation of AB

Product of $m \times n$ and $n \times p$

- The (i,j)-entry of AB is the product of the i-th row of A with the j-th column of B.

$$AB = \begin{bmatrix} \mathbf{r}_1 \\ \mathbf{r}_2 \\ \vdots \\ \mathbf{r}_m \end{bmatrix} \begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_p \end{bmatrix} = \begin{bmatrix} \mathbf{r}_1 \cdot \mathbf{c}_1 & \mathbf{r}_1 \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_1 \cdot \mathbf{c}_p \\ \mathbf{r}_2 \cdot \mathbf{c}_1 & \mathbf{r}_2 \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_2 \cdot \mathbf{c}_p \\ \vdots & \vdots & & \vdots \\ \mathbf{r}_m \cdot \mathbf{c}_1 & \mathbf{r}_m \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_m \cdot \mathbf{c}_p \end{bmatrix}$$

- In other words, if $A = [a_{ij}]$ is $m \times n$ and $B = [b_{ij}]$ is $n \times p$ then the (i,j)-entry of AB is

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

$$\begin{bmatrix} 2 & -1 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 2(0) + (-1)4 & 2(1) + (-1)(1) \\ (-2)(0) + 3(4) & (-2)(1) + 3(1) \end{bmatrix} = \begin{bmatrix} -4 & 1 \\ 12 & 1 \end{bmatrix}$$

Another method of calculation of AB

Product of $m \times n$ and $n \times p$

- The (i,j)-entry of AB is the product of the i-th row of A with the j-th column of B.

$$AB = \begin{bmatrix} \mathbf{r}_1 \\ \mathbf{r}_2 \\ \vdots \\ \mathbf{r}_m \end{bmatrix} \begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_p \end{bmatrix} = \begin{bmatrix} \mathbf{r}_1 \cdot \mathbf{c}_1 & \mathbf{r}_1 \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_1 \cdot \mathbf{c}_p \\ \mathbf{r}_2 \cdot \mathbf{c}_1 & \mathbf{r}_2 \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_2 \cdot \mathbf{c}_p \\ \vdots & \vdots & & \vdots \\ \mathbf{r}_m \cdot \mathbf{c}_1 & \mathbf{r}_m \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_m \cdot \mathbf{c}_p \end{bmatrix}$$

- In other words, if $A = [a_{ij}]$ is $m \times n$ and $B = [b_{ij}]$ is $n \times p$ then the (i,j)-entry of AB is

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

$$\begin{bmatrix} 3 & -1 \\ 2 & 5 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 & 5 \\ -2 & 1 & 1 \end{bmatrix} = \begin{bmatrix} (3)(0) + (-1)(-2) & (3)(-1) + (-1)(1) & (3)(5) + (-1)(1) \\ (2)(0) + (5)(-2) & (2)(-1) + (5)(1) & (2)(5) + (5)(1) \\ (-1)(0) + (0)(-2) & (-1)(-1) + (0)(1) & (-1)(5) + (0)(1) \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -4 & 14 \\ -10 & 3 & 15 \\ 0 & 1 & -5 \end{bmatrix}$$

Example

$$\begin{bmatrix} 1 \\ -2 \\ 0 \\ 3 \end{bmatrix} \begin{bmatrix} -2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} -2 & 3 & 4 \\ 4 & -6 & -8 \\ 0 & 0 & 0 \\ -6 & 9 & 12 \end{bmatrix}$$

However,

$$\begin{bmatrix} -2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 0 \\ 3 \end{bmatrix} = ☹$$

- Note that it is possible to have $AB = AC$ but $B \neq C$.

  Example.

$$\underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}}_{A}\underbrace{\begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}}_{B} = \begin{bmatrix} 0 & -1 \\ 0 & -2 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}}_{A}\underbrace{\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}}_{C}$$

  but

$$\begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \neq \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

- We can have $A \neq 0$ and $B \neq 0$ while $AB = 0$.

  Example.

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Example

If $A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

then

$$AB = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}$$

but

$$BA = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$$

so

$$AB \neq BA.$$

Recall that I_m denotes the $m \times m$ identity matrix, e.g., $\mathbf{I}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Suppose A is an $m \times n$ matrix. Then

$$\mathrm{I}_m A = A\mathrm{I}_n = A$$

In particular, if A is an $n \times n$ matrix, then

$$\mathrm{I}_n A = A\mathrm{I}_n = A$$

Example.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -3 & 2 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -3 & 2 \\ 1 & 0 \end{bmatrix}$$

or

$$\begin{bmatrix} 2 & 1 \\ -3 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -3 & 2 \\ 1 & 0 \end{bmatrix}$$

If $\mathbf{x}$ is a vector in $\mathbb{R}^n$ then $\mathrm{I}_n\mathbf{x} = \mathbf{x}$.

Example.

$$\mathbf{x} = \begin{bmatrix} 1 \\ -2 \\ 0 \\ -3 \end{bmatrix} \quad \Rightarrow \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 0 \\ -3 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 0 \\ -3 \end{bmatrix}$$

Powers of a matrix

Given a square matrix A and a positive integer k, we define A^k by

$$A^k = \underbrace{AA\cdots A}_{k \text{ times}}$$

- For instance $A^2 = AA$, $A^3 = AAA$, $A^4 = AAAA$, ...

Example.

Suppose $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Then $A^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$,

$A^3 = (A^2)A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$,

$A^4 = (A^3)A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$,

Question. What is A^{2018}?

Answer. From the above pattern it is easily guessed that $A^{2018} = \begin{bmatrix} 1 & 2018 \\ 0 & 1 \end{bmatrix}$

Example

Let $A = \begin{bmatrix} 0 & 1 & -2 \\ 0 & 0 & -3 \\ 0 & 0 & 0 \end{bmatrix}$. Compute A^{2018}.

Solution. We have $A^2 = \begin{bmatrix} 0 & 0 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $A^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

It follows that:

$$A^4 = A^5 = \cdots = A^{2018} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Transpose of A

Suppose A is an $m \times n$ matrix.

The transpose of A, which is denoted by A^T, is an $n \times m$ matrix whose rows are columns of A.

Example. Suppose $A = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 3 & -\frac{1}{2} \end{bmatrix}$. Then $A^T = \begin{bmatrix} 2 & 0 \\ -1 & 3 \\ 1 & -\frac{1}{2} \end{bmatrix}$.

Suppose $A = \begin{bmatrix} 1 & -2 & 2 \\ 0 & -3 & -5 \\ 0 & 1 & -1 \end{bmatrix}$. Then $A^T = \begin{bmatrix} 1 & 0 & 0 \\ -2 & -3 & 1 \\ 2 & -5 & -1 \end{bmatrix}$.

The transpose of a column vector is a row vector, and vice versa.

Example.

$$\begin{bmatrix} 1 & -3 & 4 \end{bmatrix}^T = \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix}$$

and

$$\begin{bmatrix} 1 \\ 0 \\ -4 \\ \frac{1}{2} \end{bmatrix}^T = \begin{bmatrix} 1 & 0 & -4 & \frac{1}{2} \end{bmatrix}$$

Basic Properties of Transpose

Elementary properties of transpose

Suppose A and B are both $m \times n$ matrices and c is a scalar. Then :

- $(A^T)^T = A$
- $(A + B)^T = A^T + B^T$ and $(cA)^T = cA^T$.

Transpose (cont.)

Suppose that A is $m \times n$ and B is $n \times p$. Then

$$(AB)^T = B^T A^T$$

Example. Suppose $A = \begin{bmatrix} 1 & -1 \\ 0 & 2 \\ 1 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 3 & 0 \\ 1 & 1 & 1 \end{bmatrix}$. Then :

$$AB = \begin{bmatrix} 1 & -1 \\ 0 & 2 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} -2 & 3 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} -3 & 2 & -1 \\ 2 & 2 & 2 \\ -4 & 1 & -2 \end{bmatrix}$$

hence $(AB)^T = \begin{bmatrix} -3 & 2 & -4 \\ 2 & 2 & 1 \\ -1 & 2 & -2 \end{bmatrix}$ and we also have:

$$B^T A^T = \begin{bmatrix} -2 & 1 \\ 3 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ -1 & 2 & -2 \end{bmatrix} = \begin{bmatrix} -3 & 2 & -4 \\ 2 & 2 & 1 \\ -1 & 2 & -2 \end{bmatrix}$$

Remark. Note that in general $(AB)^T \neq A^T B^T$. In fact, $A^T B^T$ might not necessarily be defined!

For example, if A is 2×3 and B is 3×5, then $(AB)^T$ is defined and is a 5×2 matrix, whereas $A^T B^T$ is not even defined!

$$\underbrace{\begin{bmatrix} * & * & * \\ * & * & * \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{bmatrix}}_{B} = \underbrace{\begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \end{bmatrix}}_{AB}$$

$$\underbrace{\begin{bmatrix} * & * \\ * & * \\ * & * \end{bmatrix}}_{A^T} \underbrace{\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix}}_{B^T} = \text{☹}$$

Transpose (cont.)

For more than two matrices, the formula is similar:

If A is $m \times n$, B is $n \times p$, and C is $p \times q$, then

$$(ABC)^T = C^T B^T A^T$$

.

A similar formula holds for four or even more matrices:

$$(A_1 A_2 \cdots A_{k-1} A_k)^T = A_k^T A_{k-1}^T \cdots A_2^T A_1^T$$

Inverse of a matrix

- Recall that the inverse of a number $c \neq 0$ is just $\frac{1}{c}$ or c^{-1}, and we know that $c \cdot \frac{1}{c} = \frac{1}{c} \cdot c = 1$.
- We will generalize this notion to square matrices.

Suppose A is an $n \times n$ matrix. We say that A is invertible if there exists an $n \times n$ matrix C such that

$$AC = CA = \mathrm{I}_n$$

The matrix C is called the inverse of A, and is usually denoted by A^{-1}.

Example. Suppose $A = \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}$. Then if we take $C = \begin{bmatrix} -1 & 3 \\ 1 & -2 \end{bmatrix}$ we will have

$$AC = \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} -1 & 3 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad CA = \begin{bmatrix} -1 & 3 \\ 1 & -2 \end{bmatrix}\begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

This means that A is invertible and C is the inverse of A.

Uniqueness of inverse

- If an $n \times n$ matrix A is invertible, then its inverse is unique.

  Reason : If $AC = CA = \mathrm{I}_n$ and $AD = DA = \mathrm{I}_n$, then:

$$C = C\mathrm{I}_n = C(AD) = (CA)D = \mathrm{I}_n D = D.$$

Negative powers of a matrix

If a square matrix A is invertible, we can define its negative powers: for every positive integer k,

$$A^{-k} = \underbrace{A^{-1}A^{-1}\cdots A^{-1}}_{k \text{ times}}$$

Cancellation law

Suppose that A is an invertible $n \times n$ matrix, and B and C are arbitrary $n \times p$ matrices. If $AB = AC$ then $B = C$.

Reason: $AB = AC \Rightarrow A^{-1}AB = A^{-1}AC \Rightarrow \mathrm{I}_n B = \mathrm{I}_n C \Rightarrow B = C$.

Some square matrices do not have an inverse.

Example. The matrix $A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ does not have an inverse, because for every 2×2 matrix C we have $AC = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \neq \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Example. Consider the matrix $A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$.Then for every 2×2 matrix $C = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ we have

$$AC = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ c & d \end{bmatrix}$$

The first row of AC is always zero. Therefore we cannot find a matrix C for which $AC = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. This means that A *does not have an inverse*.

Question. When does an $n \times n$ matrix A have an inverse?

- We will answer this question in this lecture and the next lecture.
- We start by a special case : when A is a 2×2 matrix.

Theorem

Suppose $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then :

- If $ad - bc = 0$ then A is **not** invertible.
- If $ad - bc \neq 0$ then A is invertible and we have

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Example. The matrix $A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$ is not invertible because $1 \times 6 - 3 \times 2 = 6 - 6 = 0$.

Example. The matrix $A = \begin{bmatrix} 1 & 3 \\ 2 & 2 \end{bmatrix}$ is invertible because $1 \times 2 - 3 \times 2 = -4 \neq 0$.

Moreover,

$$A^{-1} = \frac{1}{-4} \begin{bmatrix} 2 & -3 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & \frac{3}{4} \\ \frac{1}{2} & -\frac{1}{4} \end{bmatrix} \Rightarrow AA^{-1} = A^{-1}A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Theorem

Suppose $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then

- If $ad - bc = 0$ then A is not invertible.
- If $ad - bc \neq 0$ then A is invertible and we have

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Reason :

$$A^{-1}A = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \frac{1}{ad - bc} \begin{bmatrix} ad - bc & 0 \\ 0 & ad - bc \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The calculation for AA^{-1} is very similar.

- The quantity $ad - bc$ is called the determinant of the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Example

Is $A = \begin{bmatrix} 4 & 3 \\ -\frac{1}{3} & 1 \end{bmatrix}$ invertible? If so, then compute A^{-1}.

Solution. Determinant of A is $(4 \times 1) - (3 \times (-\frac{1}{3})) = 4 - (-1) = 5 \neq 0$. Therefore A is invertible and

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 1 & -3 \\ \frac{1}{3} & 4 \end{bmatrix} = \begin{bmatrix} \frac{1}{5} & -\frac{3}{5} \\ \frac{1}{15} & \frac{4}{5} \end{bmatrix}$$

Then,

$$AA^{-1} = \begin{bmatrix} 4 & 3 \\ -\frac{1}{3} & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{5} & -\frac{3}{5} \\ \frac{1}{15} & \frac{4}{5} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

and

$$A^{-1}A = \begin{bmatrix} \frac{1}{5} & -\frac{3}{5} \\ \frac{1}{15} & \frac{4}{5} \end{bmatrix} \begin{bmatrix} 4 & 3 \\ -\frac{1}{3} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Application of matrix inverse

Theorem

Consider a matrix equation $A\mathbf{x} = \mathbf{b}$. If A is a square $n \times n$ matrix which is invertible, then we can solve the system as follows :

$$A\mathbf{x} = \mathbf{b} \Rightarrow A^{-1}A\mathbf{x} = A^{-1}\mathbf{b} \Rightarrow \mathrm{I}_n\mathbf{x} = A^{-1}\mathbf{b} \Rightarrow \mathbf{x} = A^{-1}\mathbf{b}.$$

Therefore the solution to the above equation is $\mathbf{x} = A^{-1}\mathbf{b}$.

Example. Solve the matrix equation $\begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix}\mathbf{x} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$.

Solution. $A = \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix} \Rightarrow \det(A) = 3 \neq 0 \Rightarrow A^{-1} = \frac{1}{3}\begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}$ and

$$\mathbf{x} = \frac{1}{3}\begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

- The new method of solving $A\mathbf{x} = \mathbf{b}$ is only applicable when we know A is a square matrix which is invertible.
- Even if A is square and invertible, calculating the inverse of A can be harder than the usual method of solving a linear system.
- However, in some applications we might need to solve $A\mathbf{x} = \mathbf{b}$ for a fixed A but for several different $\mathbf{b}$. In such cases, the new method is better because it avoids the calculation of the R.E.F. of $[A|\mathbf{b}]$ over and over again.